The Positive-Chamber Koide Identity for Icosahedral Face Circulants

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Abstract

The stabilizer of an oriented icosahedral face acts regularly on its three corners. Every Hermitian response commuting with this cyclic action is a circulant $C = aI + bR + \bar{b}R^2$, $R^3 = I$. In the chamber where all three eigenvalues are nonnegative and may be read as square-root masses, its Koide invariant is

$$Q = \frac{1}{3} + \frac{2}{3}\left(\frac{|b|}{a}\right)^2.$$

Thus $Q = 2/3$ is exactly equivalent to the single balance condition $|b|/a = 1/\sqrt{2}$. This identity is a mathematical result about positive C_3 face circulants. The phase of b drops out of Q and controls the two mass ratios jointly; it is supplied, not derived. A finite tracial Gelfand–Naimark–Segal construction that pairs the neutral line with a two-state orientation record and the charged plane with one oriented event gives the balanced modulus as a conditional theorem. The balance itself is a target-informed premise: it is chosen from the measured charged-lepton triple and is algebraically the relation Koide published. Under the balance and mass-ordering premises the measured electron and muon masses fix the tau mass inside a 72-eV enclosure centered at 1776.969027 MeV. The world average lies 0.43 and Belle II 2023 lies 0.89 of their standard uncertainties away; both comparisons are retrospective, and a registered decision rule with a 0.045 MeV precision floor governs later measurements. The local face fiber has no constructed physical chiral-family attachment, so the result is a representation-theoretic identity, a finite conditional implication and a conditional tau-mass window; no source-only mass prediction is claimed.

Keywords: Koide relation, circulant matrices, cyclic symmetry, icosahedral symmetry, finite event algebra, GNS representation

1 The face-corner carrier

Koide's charged-lepton relation [1] is usually written in terms of the three positive square-root masses. The construction below identifies the exact condition it imposes on a cyclic Hermitian response.

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Let $G = A_5$, the alternating group on five letters, act by proper rotations on the icosahedron. The twenty outward oriented faces form the transitive orbit G/C_3. The stabilizer of one face cyclically permutes its three corners, so every face carries a local copy of the regular C_3 representation. If R denotes the cyclic shift, then

$$R^3 = I, \qquad R^\dagger = R^2.$$

The commutant of this regular action is the three-dimensional circulant algebra

$$\{x_0 I + x_1 R + x_2 R^2 : x_j \in \mathbb{C}\}.$$

Hermiticity restricts a response to

$$C = aI + bR + \bar{b}R^2, \qquad a \in \mathbb{R}, \quad b \in \mathbb{C}. \tag{1}$$

Write $b = \rho e^{i\delta}$, with $\rho = |b| \geq 0$. Fourier diagonalization gives

$$\lambda_k = a + 2\rho \cos\left(\delta + \frac{2\pi k}{3}\right), \qquad k = 0, 1, 2. \tag{2}$$

The unordered spectrum is independent of the chosen face representative. This is a local bundle statement. The sixty face-corner flags form the regular A_5 torsor; the geometry alone supplies no canonical global three-dimensional physical family space.

2 The positive-chamber identity

Theorem 1 (Positive-chamber Koide identity)**.** *Let C be the Hermitian circulant in Eq. (1), with $a > 0$. Suppose its three eigenvalues in Eq. (2) are nonnegative and, for one $s > 0$,*

$$\sqrt{m_k} = \sqrt{s}\lambda_k.$$

Then

$$Q := \frac{\sum_{k=0}^{2} m_k}{\left(\sum_{k=0}^{2} \sqrt{m_k}\right)^2} = \frac{1}{3} + \frac{2}{3}\left(\frac{\rho}{a}\right)^2. \tag{3}$$

Consequently,

$$Q = \frac{2}{3} \quad \Longleftrightarrow \quad \frac{\rho}{a} = \frac{1}{\sqrt{2}}. \tag{4}$$

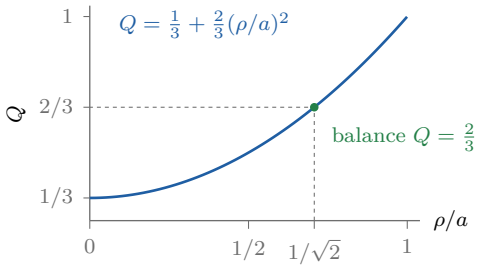
Proof. Set $c_k = \cos(\delta + 2\pi k/3)$. The roots-of-unity identities give

$$\sum_{k=0}^{2} c_k = 0, \qquad \sum_{k=0}^{2} c_k^2 = \frac{3}{2}.$$

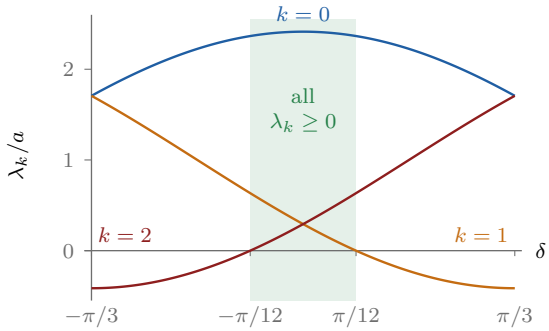
It follows that

$$\sum_k \lambda_k = 3a, \qquad \sum_k \lambda_k^2 = 3a^2 + 6\rho^2.$$

The scale s cancels from Q, yielding Eq. (3). Since $\rho/a \geq 0$, Eq. (3) equals $2/3$ exactly at $\rho/a = 1/\sqrt{2}$. □



(a) Koide ratio when $\sqrt{m_k} \propto \lambda_k \geq 0$



(b) $\lambda_k/a = 1 + \sqrt{2}\cos(\delta + 2\pi k/3)$ at balance; the positive chamber is $|\delta| \leq \pi/12$ modulo $2\pi/3$

Figure 1: (a) The Koide ratio of a nonnegative spectrum of $C = aI + bR + \bar{b}R^2$, $\rho = |b|$, as a function of ρ/a (Theorem 1); balance $\rho/a = 1/\sqrt{2}$ gives $Q = 2/3$. (b) At balance the three eigenvalues are nonnegative exactly in the chamber $|\delta| \leq \pi/12$ modulo $2\pi/3$, the chamber outside which the theorem makes no statement.

Remark 2 (The chamber is part of the theorem). The signed trace calculation holds algebraically for every δ. Physical square roots use $|\lambda_k|$ when an eigenvalue is negative, and then the denominator is not the signed trace. At $\rho/a = 1/\sqrt{2}$, the positive chamber is

$$|\delta| \leq \frac{\pi}{12} \pmod{\frac{2\pi}{3}}.$$

Theorem 1 makes no physical Koide statement outside that chamber.

Remark 3 (What the phase contains). Inside the positive chamber, Q is independent of δ. The three eigenvalues, and hence the two reported mass ratios, vary with δ. The balance condition removes one scale-free degree of freedom from a three-mass spectrum. It does not determine the ratios.

3 Conditional finite tracial balance

The balance condition in Eq. (4) can arise exactly from a finite event packet. This section states the packet as an independent conditional theorem. The packet realizes the target-informed balance premise through an asymmetric block choice, which is where that premise enters.

Let

$$\mathcal{V} = \mathbf{1} \oplus \chi \oplus \overline{\chi}, \qquad \mathcal{H}_{\mathrm{or}} = \mathbb{C}^2, \qquad \mathcal{A} = B(\mathcal{V} \otimes \mathcal{H}_{\mathrm{or}}) \simeq M_6(\mathbb{C}).$$

Here $\mathbf{1}$ is the neutral cyclic mode, $\chi \oplus \overline{\chi}$ is the two-dimensional charged plane, and $\mathcal{H}_{\mathrm{or}}$ is a two-state orientation record. Let P_0 and P_c project onto the neutral line and charged plane. For a rank-one oriented event q_+, define

$$Z_0 = P_0 \otimes I_{\mathrm{or}}, \qquad Z_c = P_c \otimes q_+, \qquad E_+ = Z_0 + Z_c.$$

Both Z_0 and Z_c have rank two.

Theorem 4 (Conditional finite tracial-GNS balance)**.** *Condition the normalized trace state $I_6/6$ on E_+. Map the orthonormal cyclic basis of the resulting square-root amplitude to (I, R, R^2) in $L^2(M_3(\mathbb{C}), \tau_3)$. Then*

$$p_0 = p_c = \frac{1}{2}, \qquad a = \sqrt{p_0}, \qquad \rho = \sqrt{\frac{p_c}{2}},$$

and therefore

$$\frac{\rho}{a} = \frac{1}{\sqrt{2}}, \qquad Q = \frac{2}{3}$$

whenever the positive square-root-mass reading of Theorem 1 is supplied.

Proof. Normalized-trace conditioning assigns probability proportional to block rank. Both accepted blocks have rank two, so $p_0 = p_c = 1/2$. In $L^2(M_3(\mathbb{C}), \tau_3)$, the operators I, R, R^2 are orthonormal. The canonical square-root vector therefore has neutral coefficient $\sqrt{p_0}$ and two charged coefficients of modulus $\sqrt{p_c/2}$. Their ratio is $1/\sqrt{2}$. Theorem 1 then gives $Q = 2/3$ on the positive chamber. □

Theorem 4 is finite and exact. Its event algebra, block choice, conditioned trace, and cyclic response map are premises. The block choice is asymmetric: the neutral line is paired with the full two-state orientation record and the charged plane with one oriented event. Treating the two blocks alike, both with I_{or} or both with q_+, gives $p_c/p_0 = 2$, hence $\rho/a = 1$ and $Q = 1$. The normalized trace fixes the probabilities only after the asymmetric event has been chosen, so the balanced modulus is the content of that choice. A physical charged-lepton statement additionally requires a quotient-visible map from the local face bundle to one chiral three-family response, preservation of the relevant block powers, and a mass readout. Those objects are absent from the theorem. The executable observer construction supplies none of them: the tau window consumes no native observer run, and its identification of balanced square-root masses with charged-lepton pole masses is a premise [8, 6].

Remark 5 (What a physical attachment would add)*.* The connected register fixes the normalization left free by a direct-sum maximum-entropy model; minimality of this response-local register is a declared hypothesis. A physical charged-lepton conclusion requires a chiral regular-C_3 carrier and faithful trace-preserving unital completely positive maps Φ, Ψ between the source and physical response algebras, with $\Psi\Phi = \mathrm{id}$. If they intertwine the block records and identify the physical GNS vector with $C = (Y_e^\dagger Y_e)^{1/4}$, Kadison–Schwarz makes Φ a $*$-monomorphism and an exact L^2 isometry, so it carries the equal source block powers to the physical response. For an accepted finite chiral checkpoint with three left and three right family modes, positive kinetic metrics, and a committed neutral Higgs direction of electroweak scale v, $\Phi = \mathrm{Ad}_{J_L \oplus J_E}$. With $M_F = X_F^2$, this gives

$$\widehat{Y}_e = \frac{\sqrt{2}}{v} J_L M_F J_E^\dagger, \qquad \mathcal{M}_L = J_L M_F J_L^\dagger.$$

An extended branch that adds graded physical completion and quotient source-law selection selects one dressed mass readout, given an exhaustive declared carrier class with a positive winner gap, a source-closed BV/BRST continuum, and an interval or contraction certificate for the charged QFT self-map. A balanced C_3 fixed point then gives $Q = 2/3$; a C_3-symmetric fixed point with charged attenuation $\chi_\star$ gives

$$Q = \frac{1 + e^{-2\chi_\star}}{3}.$$

An off-plane response requires the full response operator. None of these maps, checkpoints or certificates is constructed.

4 The conditional tau-mass window

Fixing $Q = 2/3$ in Eq. (3) and inserting the measured electron and muon masses turns the Koide coordinate into one quadratic in $\sqrt{m_\tau}$. The mass-ordering premise $m_\tau > m_\mu$ selects the physical root; the excluded second root sits at 3.317 MeV, below the muon mass. Propagating the one-sigma corners of the measured (m_e, m_μ) inputs [2] with outward rounding encloses the selected root in

$$m_\tau \in [1776.968991,\ 1776.969063]\ \text{MeV}, \tag{5}$$

a 72-eV window with center 1776.969027 MeV. The center sits 0.43 standard uncertainties from the measured 1776.93 ± 0.09 MeV, and the window is three orders of magnitude narrower than that measurement uncertainty. The dedicated Belle II 2023 measurement, $1777.09 \pm 0.08_{\text{stat}} \pm 0.11_{\text{syst}}$ MeV, lies 0.12097 MeV above the center, 0.89 of its combined 0.136 MeV uncertainty [4]. That comparison is retrospective, and its systematic uncertainty alone exceeds the 0.045 MeV compatibility floor, so it carries no verdict under the registered rule [5].

The comparison is governed by a registered conditional test whose target definition and decision rule were fixed in a frozen prediction record [3]: a world-average or dedicated-measurement central value more than three standard uncertainties from 1776.969027 MeV refutes the balanced-circulant premise; compatibility requires agreement within two standard uncertainties at a measurement uncertainty of at most 0.045 MeV; every other outcome carries no verdict. The balance premise, the equal-weight condition on the two response sectors of one icosahedral face circulant, is target-informed, so the window is a conditional postdiction whose evidential weight lies in the kill direction [3, 6]. Improving tau-mass measurements test the balanced-circulant premise and Koide's relation together. Both send every allowed (m_e, m_μ) to the same m_τ, so under one measurement model their likelihood ratio is one and no precision in this observable distinguishes them; the formula contains no screen-grain coordinate P and constrains no other OPH parameter [6].

5 Numerical diagnostic and provenance

Using the Particle Data Group 2026 central masses [2],

$$(m_e, m_\mu, m_\tau) = (0.51099895069,\ 105.6583755,\ 1776.93)\ \text{MeV},$$

gives

$$Q_{\text{PDG}} = 0.6666644634026367, \qquad \rho/a = 0.7071044442682813, \qquad \delta = 0.2222247618945926.$$

The declared minimal complete public response (MCPR) law attenuates the balance and shifts the phase through three coefficients of a declared unified coupling u and pixel coordinate P:

$$\kappa = \frac{u/50}{1 + u/31 + u^2/310}, \qquad \chi = \frac{u^2/512}{1 - u/77}, \qquad \zeta = \frac{(P - \varphi)^2/(21\pi)}{1 - u/27 - u^2/135},$$

with $\varphi = (1 + \sqrt{5})/2$,

$$\frac{\rho}{a} = \frac{e^{-\chi}}{\sqrt{2}}, \qquad \delta = \frac{2}{9} + \zeta,$$

and a common multiplier chosen so that the product of the three masses divided by the cube of a declared electroweak scale v equals $e^{3\kappa}/(2 \cdot 6^{14})$. Inside the positive chamber, Eq. (3) gives the exact circulant identity

$$Q = \frac{1 + e^{-2\chi}}{3} \tag{6}$$

at every phase. The comparison packet of the particle construction [7] has

$$\begin{aligned} \chi &= 3.304899678958019 \times 10^{-6}, \qquad \zeta = 2.540957613468169 \times 10^{-6}, \\ Q_{\mathrm{MCPR}} &= 0.6666644634074956, \\ \rho/a &= 0.7071044442734351, \\ \frac{\rho/a - 1/\sqrt{2}}{1/\sqrt{2}} &= -3.3049 \times 10^{-6}. \end{aligned}$$

Its constants $50, 31, 310, 512, 77, 21, 27, 135$, amplitude, phase and determinant exponent are target-informed inputs, so the response is neither blind nor source-derived. The proximity of the two displayed Q values carries no significance or predictive weight.

Reading the three outputs, in the external energy unit E_*, as charged-lepton pole masses is an adopted physical premise, which similarity of dimensionless circulant coordinates does not derive. Under it the discrepancies are about -70.5 parts per million, almost entirely a common scale, and the muon/electron ratio differs by $+0.0725$ parts per million. The exact point $\rho/a = 1/\sqrt{2}$, $\delta = 2/9$ of the same circulant gives a muon/electron discrepancy of $+9.83$ parts per million. The particle paper gives the full comparison table, in which all twelve nonzero outputs lie outside their printed intervals, and no measured particle mass is derived [7].

Statement	Scope
Eq. (3)	exact algebra in the nonnegative-eigenvalue chamber
Theorem 4	exact implication conditional on the declared finite event packet
Icosahedral face fiber	exact local A_5/C_3 bundle and unordered spectrum
Tau window, Eq. (5)	exact quadratic consequence of the balance and ordering premises; registered decision rule
Physical family attachment	absent from the theorems; a premise of any physical reading
Phase and numerical ratios	balance leaves one phase controlling both scale-free ratios
Eq. (6)	exact circulant identity in the chamber
MCPR numerical proximity	target-informed comparison-only diagnostic
Charged pole comparison	adopted premise; outputs outside their printed intervals

6 Why residual symmetry does not finish the ratios

The family-shape problem can be stated in the real five-dimensional representation

$$W_5 \simeq \mathrm{Sym}^2_0(\mathbb{R}^3),$$

the traceless symmetric 3×3 matrices with action $A \mapsto gAg^T$.

Proposition 6 (Residual-stabilizer boundary)**.** *Invariance under a threefold or fivefold rotation about an axis n forces*

$$A = \alpha(nn^T - I/3),$$

which has a double eigenvalue. Invariance under a twofold rotation has a three-dimensional linear fixed locus, hence two projective parameters, and that locus admits simple spectrum.

Proof. Choose the rotation axis as the third coordinate. Decompose a traceless symmetric matrix into a planar scalar, planar spin-two anisotropy, transverse vector, and axial entry. Rotation by θ acts on the anisotropy by 2θ and on the transverse vector by θ. For $\theta = 2\pi/3$ or $2\pi/5$, neither block has a nonzero fixed vector, so only the axial combination remains. For $\theta = \pi$, the planar symmetric block survives:

$$A = \begin{pmatrix} u & v & 0 \\ v & w & 0 \\ 0 & 0 & -u-w \end{pmatrix}.$$

This space has dimension three and contains matrices with three distinct eigenvalues. □

Threefold and fivefold symmetry cannot produce three distinct masses. Twofold symmetry leaves two scale-free coordinates, exactly enough to carry the two mass ratios without selecting them. A numerical spectrum therefore requires a specific invariant potential derived from the screen dynamics. This proposition locates the missing input; it supplies no replacement fit.

7 Formal proof boundary

The algebraic core has a Lean 4 formalization. It proves the root sum, square sum, quotient formula, reciprocal-square-root identity, and the nonnegative-modulus equivalence

$$\frac{1}{3} + \frac{2}{3}r^2 = \frac{2}{3} \quad \Longleftrightarrow \quad r = \frac{1}{\sqrt{2}}.$$

The formal module deliberately does not identify signed eigenvalues with physical square roots. Positivity of all three eigenvalues, the phase selection, and the source-to-charged-family attachment are outside its statement. This division matches Theorem 1: the checked algebra is exact, and the physical reading is a separate premise.

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